

1. Stakeholder Groups

The first group is Banking Customers and Vulnerable Account Holders, who are the end users. They are individual account holders who submit complaints through banking apps, web forms, emails, and call centers about financial disputes, service issues, or billing errors. Their core priorities are rapid, fair, and accurate dispute resolution; being understood when they express issues in informal language, emotional phrasing, or local syntax; and avoiding financial loss caused by misrouted or delayed grievances, especially in urgent matters like fraud.

The second group is Customer Service Representatives (CSRs) and Complaint Resolution Teams, who act as domain experts and intermediaries. These frontline banking staff and specialized dispute handlers receive categorized complaints and carry out the actual operational resolutions. Their core priorities are reducing manual search and triage time across high volumes of incoming tickets, preventing the deskilling or displacement of their professional judgment, maintaining oversight of high-risk cases, and getting transparent, explainable model predictions with confidence scores.

2. Potential Benefits and Risks

For Banking Customers and Vulnerable Account Holders, the benefits are faster triage and shorter waiting times for dispute resolution, reduced financial friction for low-income users who are sensitive to pending transaction errors, and equitable treatment regardless of colloquial or regional phrasing. The harms and concerns include misclassification of urgent complaints (for example, flagging an unauthorized fraud transaction as a routine billing inquiry), exclusion due to non-standard regional phrasing or non-English inputs, and a lack of recourse if an automated system delays human review. On the socio-economic side, a delayed fraud investigation causes direct financial harm. On the linguistic side, customer grievances vary widely in phrasing and tone.

For Customer Service Representatives and Resolution Teams, the benefits are a drastic reduction in repetitive manual sorting, fewer triage errors, faster routing to the correct operational silos, and the ability to focus human effort on complex cases. The harms and concerns include automation bias, where staff blindly trust incorrect model routing; fear of job displacement or erosion of operational autonomy; and reputational risk if misrouted tickets cause customer escalation. The ethical and professional consideration is to maintain clear human accountability for financial resolutions and to ensure the AI works as a decision-support aid rather than a replacement.

3. Engagement Method

Banking Customers and Account Holders will be engaged through Distributed Dialogue. This method fits because customers are geographically and socio-economically diverse and communicate across many digital and physical banking channels. It will be carried out through structured feedback channels within mobile app prototypes, post-resolution surveys, and community focus groups in local branches. This captures real-world informal phrasing,

colloquialisms, and regional English syntax across diverse demographic groups, which helps refine model inputs.

CSRs and Complaint Resolution Teams will be engaged through Participatory Design. This method fits because customer service staff hold the operational expertise needed to define correct classification taxonomies and handle edge cases. It will be carried out through co-design workshops where resolution agents help refine category definitions across the 43 operational classes, audit misclassified training samples, establish confidence thresholds for human escalation, and co-design the UI dashboard interface.

4. Influence on Project Design

Feedback from these stakeholders directly shapes the model architecture, dataset construction, and system deployment.

For dataset refinement and diversity, Distributed Dialogue with customers expands the representation of informal phrasing and regional idioms, such as African-English syntax. This directly informs edge-case collection and anonymization protocols.

For taxonomy and label quality, Participatory Design with CSRs refines class boundary definitions across the 43 target categories. It helps identify subtle differences between overlapping issues, for example distinguishing a billing dispute from an unauthorized transaction.

For safety and escalation logic, staff feedback establishes calibrated confidence thresholds using Temperature Scaling. Low-confidence predictions and high-risk classes, like fraud, are automatically routed to human agents.

For the interface and human-in-the-loop workflow, co-design ensures the web demo presents clear predicted labels and confidence scores. This positions the AI tool as an assistant that supports human judgment rather than replacing human accountability.